Calorimetry lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use an energy balance to analyze mixing and heat transfer.

SUCCESS CRITERIA - I can...

1. Identify hot and cold samples and their masses.
2. Balance energy gained and lost in an ideal model.
3. Discuss heat exchange with the container and surroundings.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “calorimetry” and “isolated” to explain your model or result. Sentence starters are optional.

calorimetry: using temperature changes to track energy transfers

isolated: exchanging no relevant quantity with surroundings in this model

heat: energy transferred because of a temperature difference

01 Warm-up: bring back an earlier idea

A 0.3 kg sample absorbs 1500 J while warming 5°C. **Find** specific heat and state a necessary assumption.

Word help: specific heat = energy per mass needed for a one-degree temperature change

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $Q_{hot} + Q_{cold} = 0$; $Q = mc(T_f - T_i)$.
Ideal isolated mixing, no phase change. Heat lost is negative; heat gained is positive. Real surroundings may exchange energy.

Worked reasoning: Two equal masses of the same liquid at 20°C and 40°C mix in an insulated container. The final 30°C makes the hot sample lose the same energy the cool sample gains.

Your turn: Draw heat-transfer arrows and show the two equal temperature changes.

Sentence starter: The ... in my model represents ... . Its label shows ... .

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Equal masses of the same liquid at 20°C and 60°C are mixed in an ideal insulated container.
Predict final temperature and justify.

Word help: justify = support your answer with a reason or evidence

Sentence starter: I predict … . My model supports this because … .



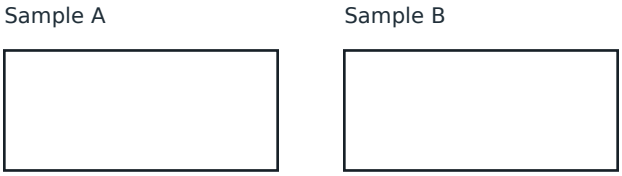
Each outline represents the labeled sample or container.

Marzano check - circle: 1 2 3

04 Apply the physics

You mix 0.10 kg of water at 80°C with 0.30 kg of water at 20°C. Ignore energy exchange with the cup and room.
Label both masses and temperatures, predict which initial temperature the final value will be closer to, then calculate equilibrium temperature.

Sentence starter: I predict … . My model supports this because … .



Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

05 Explain your reasoning

Plan a table for masses, initial temperatures, final temperature and a heat-loss observation. Why should the final temperature be measured promptly after mixing?

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

A spring has k=400 N/m and stores 0.72 J.
Find displacement magnitude and force magnitude.
Mark equilibrium, displacement and restoring-force direction.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.



Mark equilibrium and displacement x.

Marzano check - circle: 1 2 3

07 Exit check

A 0.20 kg metal cools from 100 to 40°C while 0.10 kg water warms from 20 to 40°C.
Use water c=4200. Neglect cup and losses; find metal c.

Sentence starter: I found … [unit]. This value means … in this situation.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 0.1 kg metal at 100°C is placed in 0.2 kg water at 20°C. Final temperature is 25°C; use water c=4200 and neglect container/losses.
Find metal c.

Sentence starter: I found … [unit]. This value means … in this situation.

Sample A

Sample B

Label temperatures and draw energy-transfer arrows.

Marzano check - circle: 1 2 3